



경동대학교
KYUNG DONG UNIVERSITY

Final Project Report

On

FLIGHT RESERVATION SYSTEM

By

Shema Nkindi Giscard, ID:2217151

Mukiibi Moses, ID:2217144

Umwari Deborah, ID: 2217157

Dushime Clara Cheila, ID: 2217136

Budanagi Lauria, ID: 2217148

Course: DBMS

Contents

Contents.....	2
I.INTRODUCTION	3
II.HISTORY	3
Early Developments (1930s-1960s)	3
Expansion and Advancements (1970s-1980s)	3
Modernization and Integration (1990s-2000s).....	4
Current Trends and Future Directions (2010s-Present).....	4
III. DETAILED DESCRIPTION.....	4
IV.COMPONENTS OF DATABASE DESIGN.....	5
V. RELATIONSHIPS AND CARDINALITY.....	7
VI. ENTITY RELATIONSHIP DIAGRAM.....	8
VII. RELATIONAL DATA MODEL DIAGRAM.....	10
VIII. USE CASE DIAGRAM	10
IX. SQL Queries	11
CREATE TABLE	11
INSERTING DATA INTO TABLES	12
X. REFERENCES.....	16

I.INTRODUCTION

Welcome to our Flight Reservation System, a comprehensive platform designed to simplify the process of booking air travel. Our system offers an intuitive interface that allows users to search, compare, and book flights effortlessly.

Key features include: User-Friendly Search, Real-Time Availability, Secure Booking, Personalized Options, Support and Assistance.

Our goal is to provide a hassle-free booking experience, ensuring you find the best flights at the best prices. Welcome aboard and happy travels!

II.HISTORY

The history and evolution of Flight Reservation Database Management Systems (FRDMS) are integral to the development of the broader flight reservation systems. These systems have undergone significant transformations to enhance the efficiency, accuracy, and user experience in booking flights. Here's a concise summary of its development:

Early Developments (1930s-1960s)

- **1930s-1940s:** Initially, reservations were managed manually with bookings recorded on paper. There were no dedicated database systems.
- **1950s-1960s:** The introduction of computerized systems began to change this. The SABRE system, developed by American Airlines and IBM, became operational in 1964, utilizing a centralized database to manage reservations efficiently.

Expansion and Advancements (1970s-1980s)

- **1970s:** Airlines began developing their own Computerized Reservation Systems (CRS) with robust databases to handle increasing passenger volumes. These systems could manage not just reservations but also inventory, pricing, and schedules.
- **1980s:** The establishment of Global Distribution Systems (GDS) like Amadeus, Galileo, and Worldspan, which integrated multiple airline databases, allowed travel agents to access a vast range of booking information from a single platform.

Modernization and Integration (1990s-2000s)

- **1990s:** The internet revolutionized reservation databases. Airlines and travel agencies moved their databases online, making them accessible to consumers through websites. This period saw the rise of Online Travel Agencies (OTAs) with their own integrated databases.
- **2000s:** The shift to electronic tickets (e-tickets) reduced the reliance on physical documentation, and databases became more sophisticated, managing not only reservations but also customer data, loyalty programs, and ancillary services.

Current Trends and Future Directions (2010s-Present)

- **2010s:** Mobile technology and apps allowed real-time access to reservation databases, enabling customers to book, modify, and check-in for flights from their smartphones. Advanced data analytics began to be used for personalized marketing and dynamic pricing.
- **Present and Future:** The integration of artificial intelligence (AI) and machine learning enhances the capabilities of reservation databases, allowing for predictive analytics, personalized customer experiences, and more efficient inventory management. Blockchain technology is being explored to enhance security and transparency in booking transactions.

III. DETAILED DESCRIPTION

The flight reservation system is a large scale project which includes database of more than a single Airport. It contains the details about: Airline, Passengers, Airport, Flight, Booking and Payment.

- The Flight reservation System keeps a record of its passengers:
 - Every passenger has a unique ID, name, Email, Passport number.
 - The database keeps track of transactions made by the passengers.
 - It keeps track of the booking date, and charge amount of bookings.
 - Each passenger can make the payment with many transactions.
- It also stores the information of payment with unique price when booking a ticket, changing or canceling a reservation.
 - The system retrieves payment data basing on the unique payment ID,
 - It also keeps information about the payment Method,

- Information about the amount of Payment with transaction date and time.
- Each city has an airport with a unique code and a name.
 - This database system contains details about all the passenger travelling through their respective airports.
 - The system also keeps record of the airport location and facilities
- Each Airline is identified by:
 - Its unique ID,
 - Name, operating region, and contact number
- The Flights are characterized by:
 - The flight number
 - Airline ID
 - Departure date and time
 - Arrival date and time
 - Origin Airport code
 - Destination Airport code
 - Available seats
- The systems also store data about the booking details of passengers where:
 - Each booking has a different identification number (ID),
 - It has information related to its passengers meaning that infos are revealed using the passenger ID,
 - The flight number known as Flight ID,
 - The payment status of the booking.

IV.COMPONENTS OF DATABASE DESIGN

These terminologies will help you in better understanding of the project.

An **Entity** is anything in the enterprise that is to be represented in our database. An entity can be place, person, object, event or a concept, which stores data in the database. The characteristics of entities must have an attribute, and a unique key. Every entity is made up of some 'attributes' which represent that entity.

An **Attribute** is a single-valued property of either an entity-type or a relationship type.

Primary Key is a column or group of columns in a table that uniquely identify every row in that table. These are represented in underlined form.

Foreign Key is a column that creates a relationship between two tables .The purpose of foreign keys is to maintain data integrity and allow navigation between two different instances of an entity. It is represented by an ‘*’ mark at the end of attribute.

Here we represent the database management system for an Airline.

The entities and their respective attributes required are as follows:

- **Flight**

Attributes:

- † [FlightNumber](#) (integer)
- † [DepartureDateTime](#) (date)
- † [ArrivalDateTime](#)(date)
- † [OriginAirportCode](#) (integer)
- † [DestinationAirportCode](#) (integer)
- † [AvailableSeats](#) (integer)
- † [AirlineID](#) (integer)

- **Passengers**

Attributes:

- † [FirstName](#)(text)
- † [LastName](#) (text)
- † [Email](#) (text)
- † [PassportNumber](#)(integer)
- † [PassengerID](#)(integer)

- **Booking**

Attributes:

- † [FlightID](#) (integer)
- † [PassengerID](#)(integer)
- † [PaymentStatus](#)
- † [BookingID](#) (integer)

- **Airline**

Attributes

- ✦ [AirlineName](#)(text)
- ✦ [ContactName](#) (text)
- ✦ [OpertaingRegion](#) (text)
- ✦ [AirlineID](#)(integer)

- **Airport**

Attributes

- ✦ [AirportName](#)(text)
- ✦ [Location](#)(text)
- ✦ [Facilities](#)(text)
 - ✦ [AirportCode](#) (integer)

- **Payment**

Attributes

- ✦ [BookingID](#)(integer)
- ✦ [PaymentMethod](#) (integer)
- ✦ [Amount](#) (integer)
- ✦ [TransactionDateTime](#)(Varchar)
- ✦ [PaymentID](#) (integer)

V. RELATIONSHIPS AND CARDINALITY

Relationship is nothing but an association among two or more entities.

Entities take part in relationships. We can often identify relationships with verbs or verb phrases.

Cardinality defines the numerical attributes of the relationship between two entities or entity sets. These are most useful in describing binary relation sets.

Types of relationships found in our project are described below:

Flight – Booking Relationship

Each booking is associated with one flight (FlightID in the Booking table references FlightID in the Flight table).

This is a **one-to-many** relationship, as one flight can have multiple bookings, but each booking corresponds to only one flight.

Passenger – Booking Relationship

Each booking is associated with one passenger (PassengerID in the Booking table references PassengerID in the Passenger table).

This is a **one-to-many** relationship, as one passenger can make multiple bookings, but each booking corresponds to only one passenger.

Flight – Airport Relationship

Each flight has an origin airport and a destination airport (OriginAirportCode and DestinationAirportCode in the Flight table reference AirportCode in the Airport table).

This is a **many-to-one relationship**, as multiple flights can depart from or arrive at the same airport, but each flight has only one origin and one destination airport.

Airline – Flight Relationship

Each flight is operated by one airline (AirlineID in the Flight table references AirlineID in the Airline table).

This is a **one-to-many** relationship, as one airline can operate multiple flights, but each flight is operated by only one airline.

Payment – Booking Relationship

Each payment is associated with one booking (BookingID in the Payment table references BookingID in the Booking table).

This is a **one-to-one** relationship, as each payment corresponds to only one booking, and each booking can have only one payment.

VI. ENTITY RELATIONSHIP DIAGRAM

ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities, attributes and relationships.

ER Diagrams contain different symbols, here's how they are represented

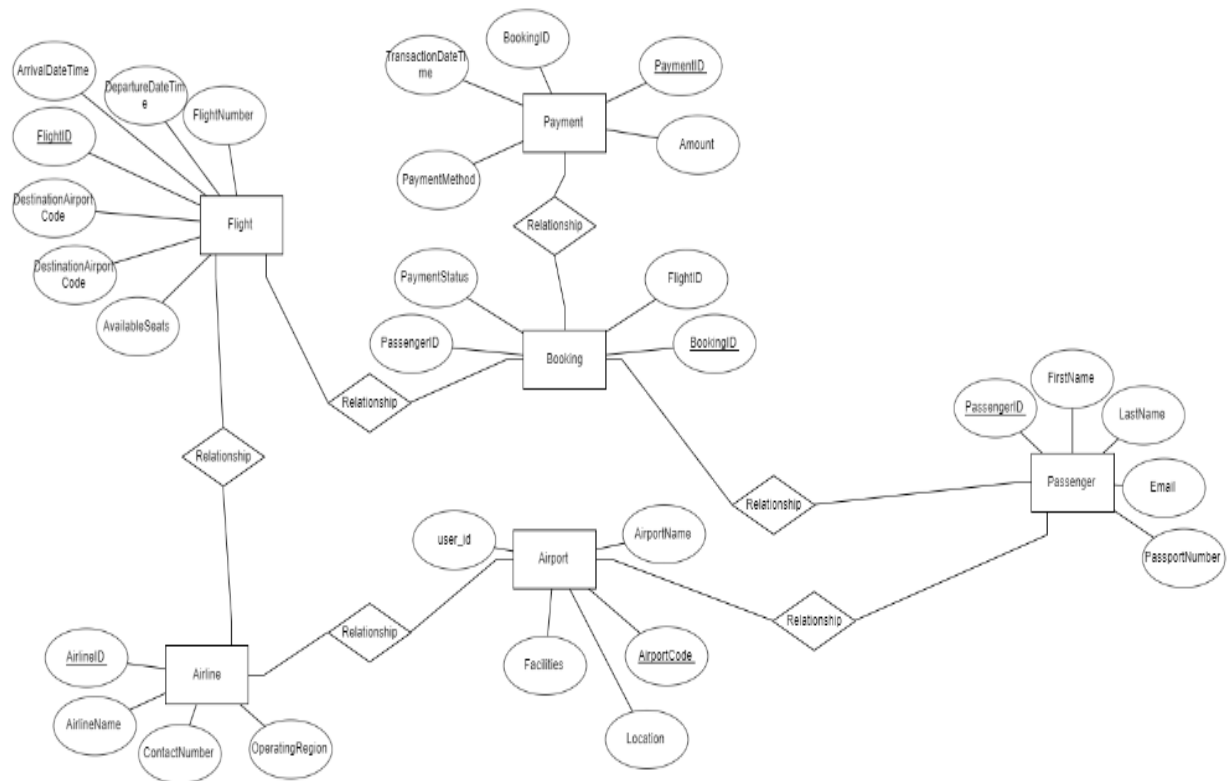
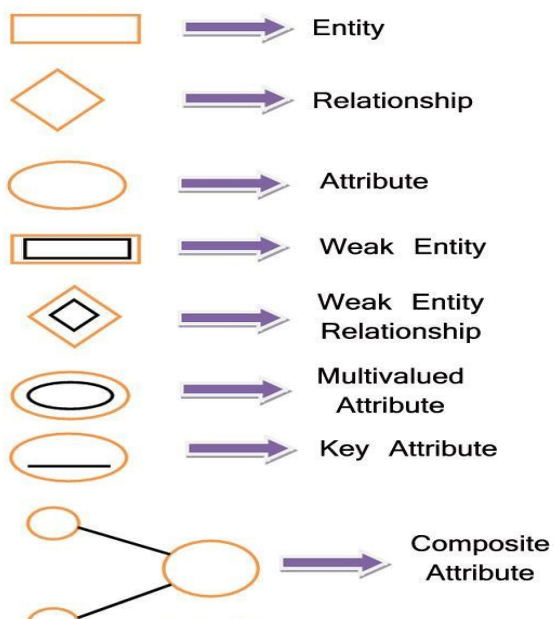


Fig 1 ERD DIAGRAM



VII. RELATIONAL DATA MODEL DIAGRAM

This is used to show the actual tables and their relationships in a database, including details like primary and foreign keys, indexes and constraints.

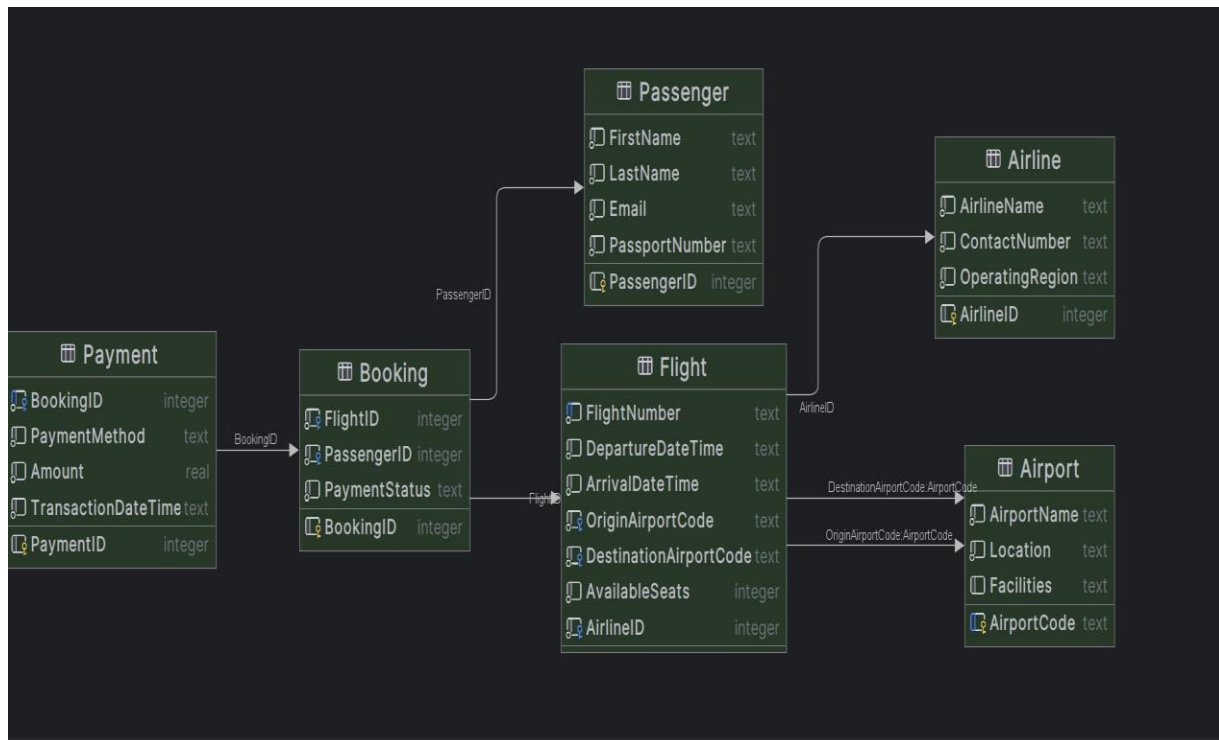


Fig 2 RELATIONAL DATA MODEL DIAGRAM

VIII. USE CASE DIAGRAM

These diagrams identify the interactions between the system and actors. The use cases and actors in use-case diagrams describe what the system does and how the actors use it, but not how the system operates internally. In short, use case diagrams describe the high-level functions and scope of a system.

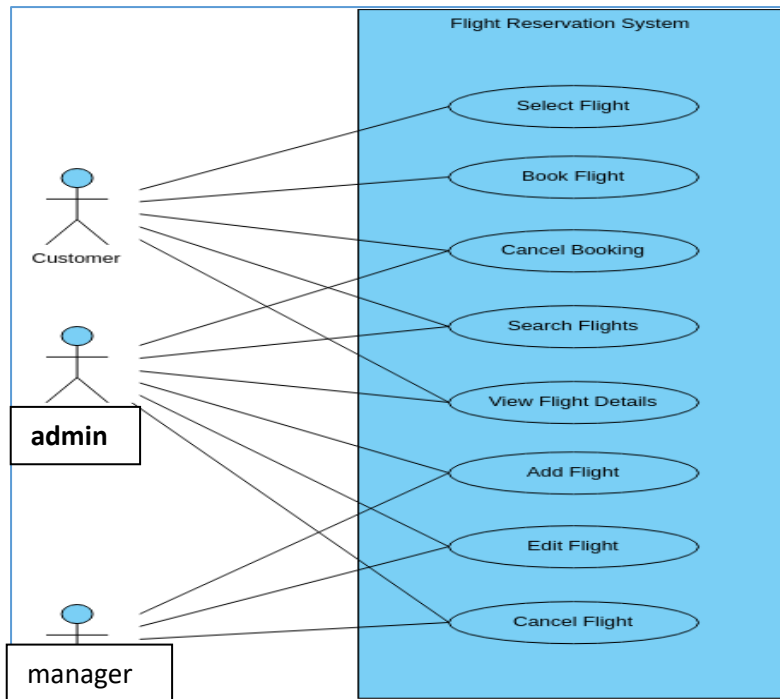


Fig 3 USE CASE DIAGRAM

IX. SQL Queries

CREATE TABLE

- Flight:**

```

CREATE TABLE IF NOT EXISTS Flight (
    FlightID INTEGER PRIMARY KEY AUTOINCREMENT,
    FlightNumber TEXT UNIQUE NOT NULL,
    DepartureDateTime TEXT NOT NULL,
    ArrivalDateTime TEXT NOT NULL,
    OriginAirportCode TEXT NOT NULL,
    DestinationAirportCode TEXT NOT NULL,
    AvailableSeats INTEGER NOT NULL,
    AirlineID INTEGER NOT NULL,
    FOREIGN KEY (OriginAirportCode) REFERENCES Airport(AirportCode),
    FOREIGN KEY (DestinationAirportCode) REFERENCES Airport(AirportCode),
    FOREIGN KEY (AirlineID) REFERENCES Airline(AirlineID)
)

```

- **Passenger:**

```
CREATE TABLE IF NOT EXISTS Passenger (  
    PassengerID INTEGER PRIMARY KEY AUTOINCREMENT,  
    FirstName TEXT NOT NULL,  
    LastName TEXT NOT NULL,  
    Email TEXT NOT NULL,  
    PassportNumber TEXT NOT NULL  
)
```

- **Airport:**

```
CREATE TABLE IF NOT EXISTS Airport (  
    AirportCode TEXT PRIMARY KEY,  
    AirportName TEXT NOT NULL,  
    Location TEXT NOT NULL,  
    Facilities TEXT)
```

- **Airline:**

```
CREATE TABLE IF NOT EXISTS Airline (  
    AirlineID INTEGER PRIMARY KEY AUTOINCREMENT,  
    AirlineName TEXT NOT NULL,  
    ContactNumber TEXT NOT NULL,  
    OperatingRegion TEXT NOT NULL)
```

- **Booking:**

```
CREATE TABLE IF NOT EXISTS Booking (  
    BookingID INTEGER PRIMARY KEY AUTOINCREMENT,  
    FlightID INTEGER NOT NULL,  
    PassengerID INTEGER NOT NULL,  
    PaymentStatus TEXT NOT NULL,  
    FOREIGN KEY (FlightID) REFERENCES Flight(FlightID),  
    FOREIGN KEY (PassengerID) REFERENCES Passenger(PassengerID))
```

- **Payment:**

```
CREATE TABLE IF NOT EXISTS Payment (  
    PaymentID INTEGER PRIMARY KEY AUTOINCREMENT,  
    BookingID INTEGER NOT NULL UNIQUE,  
    PaymentMethod TEXT NOT NULL,  
    Amount REAL NOT NULL,  
    TransactionDateTime TEXT NOT NULL,  
    FOREIGN KEY (BookingID) REFERENCES Booking(BookingID)  
)
```

INSERTING DATA INTO TABLES

- **TABLE AIRPORT:**

```
INSERT INTO Airport (AirportCode, AirportName, Location, Facilities) VALUES
(?, ?, ?, ?)", [
    ("JFK", "John F. Kennedy International Airport", "New York", "WiFi, Restaurants"),
    ("LAX", "Los Angeles International Airport", "Los Angeles", "Shopping, Lounges"),
    ("ORD", "O'Hare International Airport", "Chicago", "Parking, Duty-Free Shops"),
    ("DXB", "Dubai International Airport", "Dubai", "Shopping, Lounges"),
    ("FRA", "Frankfurt Airport", "Frankfurt", "Restaurants, Lounges")
]
```

AirportCode	AirportName	Location	Facilities
JFK	John F. Kennedy International Airport	New York	WiFi, Restaurants
LAX	Los Angeles International Airport	Los Angeles	Shopping, Lounges
ORD	O'Hare International Airport	Chicago	Parking, Duty-Free Shops
DXB	Dubai International Airport	Dubai	Shopping, Lounges
FRA	Frankfurt Airport	Frankfurt	Restaurants, Lounges

- **TABLE AIRLINE:**

```
INSERT INTO Airline (AirlineName, ContactNumber, OperatingRegion) VALUES (?, ?, ?)",
[
    ("Delta Air Lines", "123-456-7890", "North America"),
    ("American Airlines", "987-654-3210", "North America"),
    ("United Airlines", "456-789-0123", "North America"),
    ("Emirates", "971-4-123-4567", "Middle East"),
    ("Lufthansa", "49-69-123-4567", "Europe")
]
```

AirlineID	AirlineName	ContactNumber	OperatingRegion
5	Delta Air Lines	123-456-7890	North America
6	American Airlines	987-654-3210	North America
7	United Airlines	456-789-0123	North America
8	Emirates	971-4-123-4567	Middle East
9	Lufthansa	49-69-123-4567	Europe

- **Table FLIGHT:**

```
INSERT INTO Flight (FlightNumber, DepartureDateTime, ArrivalDateTime,
OriginAirportCode, DestinationAirportCode, AvailableSeats, AirlineID) VALUES
(?, ?, ?, ?, ?, ?, ?)", [
    ("DL100", "2024-06-01 08:00:00", "2024-06-01 10:30:00", "JFK", "LAX", 150, 1),
    ("AA200", "2024-06-02 10:00:00", "2024-06-02 13:30:00", "LAX", "ORD", 200, 2),
    ("UA300", "2024-06-03 12:00:00", "2024-06-03 15:30:00", "ORD", "JFK", 180, 3),
    ("EK400", "2024-06-04 14:00:00", "2024-06-04 18:30:00", "JFK", "DXB", 220, 4),
    ("LH500", "2024-06-05 16:00:00", "2024-06-05 20:30:00", "ORD", "FRA", 190, 5)
]
```

```
SELECT * FROM Booking;
SELECT * FROM Flight;
SELECT * FROM Airport WHERE AirportCode = 'JFK';
```

es

Output main.Flight x

	FlightID	FlightCode	DepartureDate	ArrivalDate	Origin	Destination	Amount	Airline	
1	24	DL100	2024-06-01 08:00:00	2024-06-01 10:...	JFK	LAX	150		1
2	25	AA200	2024-06-02 10:00:00	2024-06-02 13:...	LAX	ORD	200		2
3	26	UA300	2024-06-03 12:00:00	2024-06-03 15:...	ORD	JFK	180		3
4	27	EK400	2024-06-04 14:00:00	2024-06-04 18:...	JFK	DXB	220		4
5	28	LH500	2024-06-05 16:00:00	2024-06-05 20:...	ORD	FRA	190		5

- **Table BOOKING:**

```
INSERT INTO Booking (FlightID, PassengerID, PaymentStatus) VALUES (?, ?, ?)", [
    (1, 1, "Paid"),
    (2, 2, "Pending"),
    (3, 3, "Paid")]
```

```
SELECT * FROM Booking;
SELECT * FROM Flight;
SELECT * FROM Airport WHERE AirportCode = 'JFK';
```

Output main.Booking x

	BookingID	FlightID	PassengerID	PaymentStatus
	29		1	1 Paid
	30		2	2 Pending
	31		3	3 Paid

- **Table PAYMENT:**

```
INSERT INTO Payment (BookingID, PaymentMethod, Amount, TransactionDateTime)
VALUES (?, ?, ?, ?)", [
    (1, "Credit Card", 250.00, "2024-06-01 07:45:00"),
    (3, "PayPal", 300.00, "2024-06-03 11:30:00")]
```

```

3 SELECT * FROM Airport;
4 ✓ SELECT * FROM Payment;
5 SELECT * FROM Booking;

```

Services

Output main.Payment ×

	PaymentID	BookingID	PaymentMethod	Amount	TransactionDate...
1	11	1	Credit Card	250	2024-06-01 07:45:00
2	12	3	PayPal	300	2024-06-03 11:30:00

SQL commands for retrieving data:

- Retrieve data from Passenger table;

```

SELECT * FROM Passenger
WHERE LastName LIKE '%Smith%';

```

Services

Output main.Passenger ×

	PassengerID	FirstName	LastName	Email	PassportNumber
1	10	Jane	Smith	jane.smith@example.com	CD789012

- Retrieve data from flight table;

```

SELECT * FROM Flight WHERE FlightID = 8;

```

Services

Output main.Flight ×

	FlightID	FlightNum...	DepartureDa...	Arri...	OriginAi...	De...		AirL...
1	8	LH500	2024-06-05 16:00:00	2024-06-05 20:...	ORD	FRA	190	5

- Update some data from the table Flight;

```

UPDATE Flight
SET AvailableSeats = 100
WHERE FlightID = 8;
SELECT * FROM Flight WHERE FlightID = 8;

```

Services

Output main.Flight ×

	FL...	Fli...	Departure...	ArrivalID...	Origi...	Dest...	AvailableSe...	
1	8	LH500	2024-06-05 16:00:00	2024-06-05 20:30:00	ORD	FRA	100	5

X. REFERENCES

- American Airlines SABRE:
 - a) Copeland, D. G., & McKenney, J. L. (1988). Airline reservations systems: Lessons from history. *MIS Quarterly*, 12(3), 353-370.
 - b) American Airlines SABRE system
- Global Distribution Systems (GDS):
 - a) Buhalis, D., & Law, R. (2008). Progress in information technology and tourism management: 20 years on and 10 years after the Internet.
 - b) The state of eTourism research. *Tourism Management*, 29(4), 609-623.
 - c) Amadeus Company History
- Internet and Online Booking:
 - a) Pappas, N. (2008). The implications of the Internet for the travel industry: An investigation of the World Wide Web sites of major airlines. *Tourism Management*, 29(1), 38-47.
 - b) Expedia Company History
- Modern Reservation Systems:
 - a) Agatz, N. A., Fleischmann, M., & Van Nunen, J. A. (2008). E-fulfillment and multi-channel distribution – A review. *European Journal of Operational Research*, 187(2), 339-356.
 - b) How AI is changing the travel industry
- COVID-19 Impact and Blockchain:
 - a) Amankwah-Amoah, J., Khan, Z., & Wood, G. (2021). COVID-19 and business failures: The paradoxes of experience, scale, and scope for theory and practice. *European Management Journal*, 39(2), 179-184.
 - b) Blockchain in Travel
- General Overviews:
 - a) Reuland, R. (1998). Electronic distribution in the airline industry. *Journal of Air Transport Management*, 4(1), 53-60.
 - b) History of airline reservations systems

END.